

Assignment-1

1. Assume we have 10 US dollars & 25 Saudi Riyals. Write a Python Script to convert the total currency to Nepali Rupees. (Exchange rates: **1 US Dollar = 133.72 NRs** and **1 Saudi Riyal = 35.82 NRs**)

Currency in NRs

US Dollars: 10

Saudi Riyals: 25

Total Currency in NRs: 2232.7

2. Given the initial deposit, the annual interest rate, and the number of years that you invest your money in an investment account. Calculate the final amount of the investment using math.pow() function.

$$amount = deposit(1 + interest)^{years}$$

3. Write a program to find root of the equation $4x^2 - 21x + 25 = 0$ using the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

4. Write a program to take “gender” as input from user. If the user is male, give the message: Good Morning Sir. If the user is female, give the message: Good Morning Ma’am.
5. Write a program to take input color of road traffic signal from the user & show the message according to this table:

SIGNAL COLOR	MESSAGE
RED	Vehicle must stop
YELLOW	Vehicles should get ready to move
GREEN	Vehicles can move now

6. Write a program that takes user input day name. If the day is Monday, Tuesday, Wednesday, Thursday or Friday, then show “It’s a week day”. If the day is Saturday then show “It’s weekend”. If the day is Sunday then show “Yay! It’s a holiday”.
7. Use a **conditional (ternary) operator** for this exercise:
If the variable age is a value below 18, the value of the variable voteable should be "Too young", otherwise the value of voteable should be "Old enough".
8. Write a program to take input the marks obtained in three subjects & total marks. Compute & show the resulting percentage on your page. Take percentage & compute grade as per following table:

Percentage	Grade	Remarks
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Greater than or equal to 80	A-one	Excellent
Greater than or equal to 70	A	Good
Greater than or equal to 60	B	You need to improve
Less than 60	Fail	Sorry

Show the total marks, marks obtained, percentage, grade & remarks like:

Marks Sheet

Total Marks: 300

Marks Obtained: 219

Percentage: 73%

Grade: B

Remarks: You need to improve

9. Write a program to repeatedly print the value of the variable **num** which is input by user. Value should be decreasing by 0.5 each time, as long as x Value remains positive.

Enter a number: 5

5, 4.5, 4, 3.5, 3, 2.5, 2, 1.5, 1, 0.5, 0

10. Write a program using for loop that will iterate from 1 to 20. For each iteration, it will check if the current number is even or odd, and report that to the screen (e.g. "2 is even").

1 is Odd

2 is Even

3 is Odd

4 is Even

5 is Odd

and so on.